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Feasibility study of small Hydro/PV/Wind hybrid system for off-grid rural electrification in Ethiopia

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ABSTRACT

Ethiopia is among the least developed countries on the globe with a total access to electricity not exceeding 16%. About 85% of the population lives in places where access to electricity is less than 2%. One such a place, which is the subject of this study, is the Dejen district (10°13′24.03″N, 38°07′58″E) having a total population of 107,710 within 23 villages. About 14 villages (corresponding to 63,000 people) are found in the upper Blue Nile river gorge and far remote areas, which makes the task of their electrification via grid system very difficult. Kerosene for lighting, diesel for milling and pumping, biomass for cooking and dry cells for radio are being used in the non-electrified villages. Nothing has been done so far in developing the renewable energy resources, such as small-scale hydro, solar, and wind energy in the district.

In this work, feasibility of small-scale Hydro/PV/Wind based hybrid electric supply system to the district is studied. Six sites (two on Taba stream, one on Bechet stream, two on Muga stream and one on Suha stream) with small-scale hydropower potentials have been identified. The hydro potentials are analyzed with the help of GIS and data obtained from the Ministry of Water Resource (MoWR) of Ethiopia. Meteorological data from National Meteorological Agency (NMA) of Ethiopia and other sources, such as NASA, is used for the estimation of solar and wind energy potentials. Electric load for the basic needs of the community, such as, for lighting, radio, television, electric baker, water pumps and flour mills, is estimated. Primary schools and health posts are also considered for the community. HOMER energy is used for optimization and sensitivity analysis of the hybrid system. Taba (B) site is selected for detailed study and, as a final result, different system types and their component sizes are identified having a cost of energy less than \$0.16/kWh.

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1. Introduction

With efficient, reliable and cost-effective renewable energy resources, off-grid supply can be used as an alternative to the power supplied by diesel generator for rural electrification. However, due to intermittency nature of renewable resources, use of any particular renewable energy resource based system may lead to component over-sizing and unnecessary operational and lifecycle costs. Such limitations can be overcome by combining one or more renewable energy resources in a form of a hybrid system. Hybrid systems improve load factors and save maintenance and replacement costs, as the renewable resource components complement each other [1]. For optimal combination of different renewables, various types of hybrid systems and methods of techno-economic analysis are used. Excel based, linear programming, artificial intelligence, LINGO and HOMER are the most commonly used methods of hybrid system optimization techniques [1–5].

Sopian et al. [1] have discussed about the application of genetic algorithms in the optimization of hybrid system consisting of Pico hydropower, solar photovoltaic modules, diesel generator and battery sets. Kenfack et al. [2] have proposed a micro hydro-PV hybrid system, at Batocha, in Cameroon, using HOMER software for system optimization. Kanase-Patil et al. [3] studied off-grid electrification of seven villages in the Almora district of Uttarakhand state, India. In the study, biomass, solar, hydro and wind energy sources were considered and analyzed using LINGO and HOMER software's. Bakos [4] discussed feasibility of wind/hydro hybrid system to electrify remote islands in Greece. The system was simulated with the help of Monte Carlo simulation program. Connolly et al. [5] have done a comparative study of 68 computer tools for integration of renewable resource in various energy systems. Accordingly, HOMER is evaluated as one of the most applicable for optimization, feasibility and sensitivity analysis of both off-grid and grid connected micro power systems.

Drake and Mulugeta [6,7] studied the solar and wind potential distribution of Ethiopia. Regression coefficients of the angstrom equation (a and b) relating sunshine duration to daily solar radiation and Weibull parameters (shape factor, K, and the scale factor, c) are

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Nomenclature

ASTER-GDEM	Advanced Space-Borne Thermal Emission and Reflection Radiometer Global Digital Elevation Model
CC	cyclic charge dispatch strategy
CFL	compact fluorescent lamp
COE	cost of energy (kWh)
EEPCO	Ethiopian electric power corporation
ESRI	environmental systems research institute
FAO	world food aid organization
GDP	gross domestic product
GIS	geographical information system
HOMER	hybrid optimization and modeling of electrical renewables
LF	load follow dispatch strategy
LINGO	optimization modeling software for linear, non linear and integer programming
NASA	national aeronautics and space administration
NPC	net present cost
PDF	probability density function
PV	photo voltaic
SRTM	Shuttle Radar Topography Mission
SWERA	Solar and Wind Energy Resource Assessment
A_{site}	catchment area of the study catchment (m^2)
A_{gauge}	catchment area of the gauged reference catchment (m^2)
c	a scale parameter of Weibull distribution (m/s)
G_{SC}	the solar constant = 1367 (W/m^2)
H	monthly average daily radiation on a horizontal surface (MJ/m^2)
H_0	monthly average daily extraterrestrial radiation on a horizontal surface (MJ/m^2)
K	a constant known as shape factor of Weibull distribution
n	monthly average daily number of hours of bright sunshine
N	maximum possible daily bright sunshine hour
Q_{site}	river flow rate at the study point
Q_{gauge}	river flow rate at the gauge station
$v(z)$	wind velocity at height of Z m above ground (m/s)
V_r	wind speed at reference height (m/s)
Z	region's elevation above ground (m)
Z_0	surface roughness length (m)
Z_r	reference height (m)
δ	declination angle (°)
Φ	latitude angle (°)
ω_s	the sunset hour angle

estimated throughout the country. Bekele [8,9] determined solar and wind potentials of selected locations in Ethiopia and studied feasibility of wind/PV hybrid system to electrify 200 model families. In the study, HOMER is used for optimization and sensitivity analysis. Tamrat [10] has done a feasibility comparison of independent electrification at Dillamo and Gode sites in Ethiopia using either of the wind, solar PV or micro hydropower system. The author did not consider the possibility of combining the resources into a hybrid system and the analysis is done without the use of computer tool.

This paper is part of an ongoing research project under the supervision of Addis Ababa Institute of Technology. The main focus of the project is to assess the solar, hydro and wind power potentials available at Dejen district and to propose optimal hybrid combinations for electrification of the district inhabited by 10,500 families (average of six members per family). A total of six hydropower potential have been identified. The hydropower potential is estimated with the help of recorded data obtained from MoWR of Ethiopia and the GIS software. The wind speed data is taken from NASA. Recorded sunshine data from the NMA of Ethiopia is used to estimate the solar radiation at the site. The total community load is estimated on hourly basis and is shared to each site based on their hydropower potentials. In this paper, one of the six sites, Taba (B), is studied in detail. Diesel generator and bank of batteries are used as a means of backup and storage system.

HOMER is used for optimization and sensitivity analysis of the small Hydro/PV/Wind hybrid system. The software is a micropower design tool that can simulate and optimize stand-alone and grid-connected power systems with any combination of wind turbines, PV arrays, run-of-river hydropower, biomass power, internal combustion engine generators, micro-turbines, fuel cells, batteries, and hydrogen storage, serving both electric and thermal loads (by individual or district-heating systems). Hydro, wind, and solar data is input to HOMER. In addition, the size, cost and lifetime of wind turbine, PV module, converter, battery and diesel generator are defined. Furthermore, the installation cost, design flow rate and head of hydropower source are all input to the software. The HOMER algorithm considers each possible combination of the resources and determines the feasible combinations that can meet the required system load and constraints. In the results feasible combinations of the hybrid components are displayed according to their net present cost in ascending order. In addition, the performance of each component, cost of energy, excess and/or shortage of energy and sensitivity results can be observed from HOMER output.

2. Background

Ethiopia is a non-oil producing country with a population estimated to be over 80 million in 2007 [11]. In terms of gross domestic product (GDP) per capita, Ethiopia is ranked 174th of 179 and in terms of human development index, it is rated 169th of 177 [11]. 85% of the population lives in rural areas and the total access of electricity within the country is about 16% [12]. Most of the customers live in urban areas and consume less than 50 kWh per year, while rural area electricity coverage is negligible, about 2% [12,13].

Five years back the country has started an Accelerated Development Strategy and programs to End Poverty. For an economic growth rate of 7–10%, the resulting electric energy demand growth rate is about 17% [14]. But, Ethiopia is developing at an economic growth rate of 8–12% which indicates that the energy demand growth rate is more than 17%. Due to this high energy demand rate and the climatic impacts on the hydropower system the country has been in energy shortage for long time. Power shading of every other day had been practiced even in the capital, Addis Ababa as recently as the year 2010. The problem is worse in the rural areas far from the grid system and grid electrification of these areas is unthinkable in the near future. Furthermore, hydro dominated power plants have climatic, geological and political risks [15]. Hence, attention should be given to development and diversification of the power system with renewable energy sources which are cost effective in reaching the remote areas. Ethiopia has a lot of small hydropower, solar and wind energy potentials convenient for rural electrification [6–17]. It is conceivable that a hybrid system has the advantage of improved reliability and therefore gives better energy service when compared to any particular (wind, solar, etc.) type of stand-alone supply system. What this means is that in the absence of one type of energy (example: solar energy

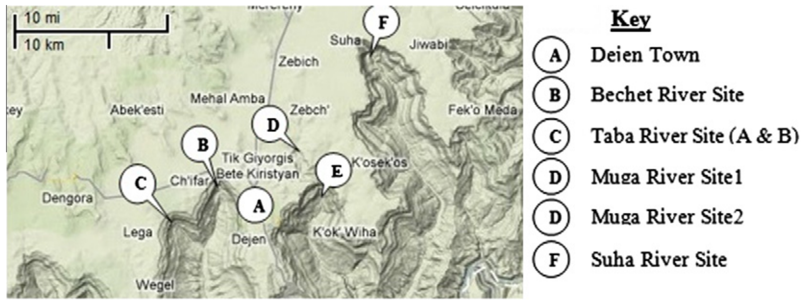

Fig. 1. Geographical layout of the project area (source: Google Map, 2010).

Fig. 2. Sample photos taken during head measurement.

Table 1
Head measurement results at the sites and villages in the neighborhood.

Site	Head (m)	Non electrified small villages in the vicinity
Taba A	50.5	Gudy, Anqraq, G/Mariam, Dengel, Muamit, Amaya, Hlgie, Chifar, Wejel, Chuhetma, Shlemuk, Mizan, Sieko
Taba B	62	
Muga 1	47	Hagere Selam, Buhach, Gafasob, Emebuga, Esmaya Dejen, Dgtma, Burabur, Buaba, Musial, Kolem, Kodem, Yeshencha, Lit't,
Muga 2	113	Gady, Enambur, Borebor, Sentayt, Gelgele, Sengrer, Worka, Alektam
Bechet	74	Dbab Zib, Yegurtina, Omarit, Byan, Ybza, Ming, Jrie, Gentay

during nighttime) another could be available (example: wind) to carry out the service. Hence, hybrid systems are found to be more appropriate than single stand-alone resources.

Dejen district is one of the rural villages without sufficient access to electricity, in spite of having significant renewable energy resources. According to the rural development office of the district, there are 23 villages in the district from which two are town villages electrified much earlier. Other four villages are on the way to be electrified. Additional three villages are in EEPCO's future electrification plan. The remaining 14 villages (about 63,000 people) may not be electrified any time soon due to their remoteness and uneasy geographical location. Before September 2010, the total electrification in the district was below 6% (including the two town villages). The aim of this paper is to study the feasibility of a hybrid system capable of electrifying those 14 villages which are not to be electrified via national grid system. The people in these villages use Kerosene for lamp, diesel for water pumping and flour mills, fire wood for cooking and dry cells for radio and tape recorders. Desertification of the land is getting worse and worse due to deforestation and backward agriculture. Fig. 1 shows the geographical layout of the project area.

Modern electricity is much cheaper, easy to handle, clean and environmental friendly compared to the use of oil and biomass. It is the main ingredient for social welfare and technological development. EEPCO, the sole authority for national electricity supply, has planned to electrify more and more villages time after time. However, due to capacity constraints and difficult geography of the remote locations, their grid connection has not been possible. On the other hand, there are tributary rivers crossing the district and joining the Blue Nile basin. Out of these Taba, Muga, Bechet and Suha are the main ones, with deep waterfalls. NASA metrological estimation for the area shows a 6.05 $kWh/m^2/day$ solar potential

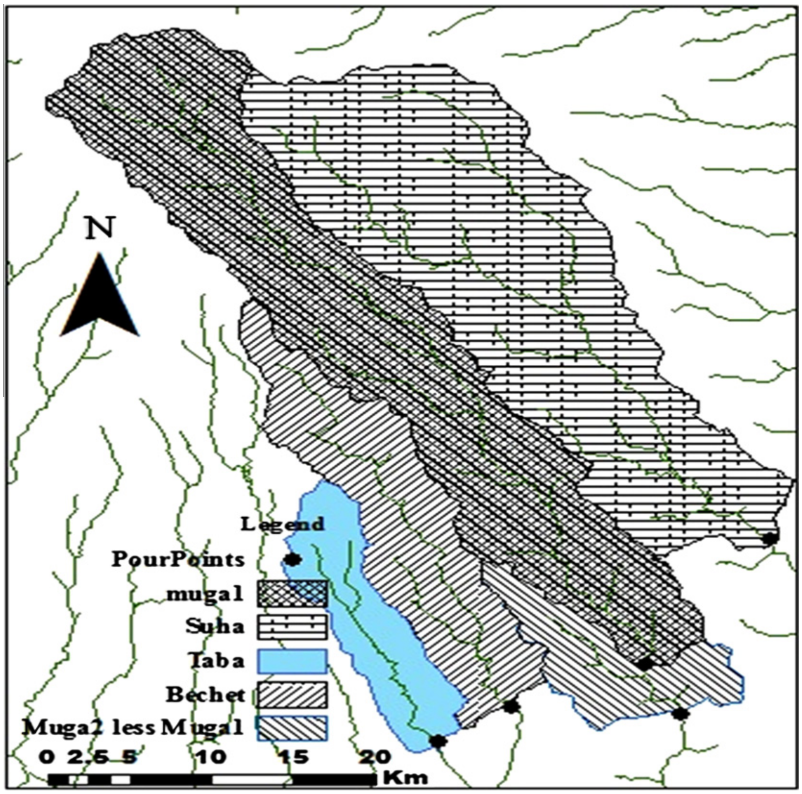

Fig. 3. Sub-basin delineation for the selected sites (pour points).

Table 2
Results of catchment extraction.

Site	Location		Catchment area (km^2)	Reference area (km^2)	$\left[\frac{A_{site}}{A_{gauge}}\right]$
	Latitude	Longitude			
Taba	10°12′25.68″N	38°3′3.00″E	93.963	375	0.25
Bechet	10°14′31.00″N	38°5′37.11″E	193.545	375	0.516
Muga1	10°16′11.05″N	38°10′15.91″E	479.58	375	1.28
Muga2	10°14′11.11″N	38°11′33.70″E	562.172	375	1.5
Suha	10°21′26.46″N	38°14′45.21″E	529.621	359	1.475

Table 3
Mean monthly flow rate (l/s) at gauged point on Muga River and at Taba site.

Month	January	February	March	April	May	June	July	August	September	October	November	December	Ave
Gauge	245	158	286	787	457	1704	24,401	30,665	6608	5704	974	734	6147
Taba	61	39	71	196	114	426	6100	7666	1652	1426	243	183	1536

and a 3.1 m/s (at 10 m height) wind speed [18]. Combining these resources into hybrid off-grid system is appropriate in such remote areas.

3. Data source and data treatment

One of the major tasks of this study is the assessment of the potentials of hydropower, wind, and solar resources, which is then followed by load estimation and feasibility analysis of the hybrid systems. Site identification and site reconnaissance is carried out with the help of Google Earth, Google Map, interviewing the local community and through repeated visits to the site. As a result, the six potential sites, (two on Taba stream, one on Bechet stream, one on Suha stream and two on Muga stream) also called pour points are identified.

3.1. The hydropower potential assessment

The proposed hydropower system is runoff the river type which requires the determination of available head and flow rate at the pour points. Head can be measured using either of altimeter, pressure gauges, clear hose method, satellite images, sighting meter or level method [19]. In the case of this study, the selected sites have steep waterfalls so that their head is measured using rope (see Fig. 2). Table 1 summarizes head measurement of each site and non electrified villages in the neighborhood.

Although there are gauge stations at Muga and Suha streams, there are no gauging stations at the selected sites. The flow rate at the point of interest is estimated using hydrological estimation techniques. According to World Meteorological Organization (WMO), the flow rate at ungauged sites can be estimated using one of the three methods [20] – Empirical, Statistical and Rainfall-Runoff modeling.

Regional methods and rainfall runoff models require plenty of data which are not available for the studied areas. All the five catchments lie in the same Northern Gojjam sub-basin of upper Blue Nile [21,22]. They are located close to each other and have similar landscape, soil type and land coverage. The catchments start from the same source known as mount Choke. Data from the FAO soil map also shows the similarity of the soil type and vegetation coverage of all the sites. Because of such geographical similarity the simple empirical estimation methods are adopted instead of the more complex statistical and rainfall runoff [19–23]. The gauge stations at Muga and Suha streams are taken as reference for empirical formulation given by Eq. (1) [19,20].

$$Q_{site} = K\left[\frac{A_{site}}{A_{gauge}}\right]Q_{gauge} \tag{1}$$

where A_{site} is the catchment area of the power plant site (m^2); A_{guage} the catchment area of the gauge (m^2); Q_{site} the discharge at site (m^3/s); Q_{guage} the discharge at gauge (m^3/s) and K is the represents a scaling constant or function.

Thus, to determine the flow rate at the pour points, the catchment area contributing to the volume flow should be known. ArcGIS 9.3 Desktop is used together with Digital Elevation Model (DEM) data for catchment area extraction [23]. DEM is a digital representation of a given geographical feature in the form of matrices. Worldwide DEM data can be downloaded free of charge from the two well-known sources. These are SRTM (Shuttle Radar Topography Mission: http://srtm.csi.cgiar.org) and ASTER-GDEM (Advanced Space-borne Thermal Emission and Reflection Radiometer Global Digital Elevation Model: http://www.ersdac.or.jp/GDEM/E/index.html). Hydrology Toolsets released by Environmental Systems Research Institute, Inc. (ESRI) with the ArcGIS 9.3 are applied on ASTER-GDEM data sources, ASTGTM_N10E37 and ASTGGTM_N10E38. Mosaic, fill, flow direction, flow accumulation, raster calculator, stream link, stream order, pour point creator, water shading and contour tools are applied sequentially to determine the catchment area of each hydropower sites and the result is depicted in Fig. 3. Each catchment area is converted to polygon feature to be able to calculate the area. Table 2 shows the result of area measurement for each site. Taba (A) and Taba (B) are very close to one another and for simplicity the same catchment area is considered.

Based on their geographical locations, gauge station at Muga River is taken as a reference catchment for Muga site 1, Muga Site 2, Bechet Site and Taba site. The gauged catchment at Suha River is taken as a reference for Suha site. The value of the catchment area ratio of each site to its corresponding reference catchment is given in Table 2. The value of K is taken to be 1 by assuming the same rainfall distribution over the sites. The scope of this paper is limited to the analysis of a hybrid system at Taba (B) site having a 62 m head. A 25 years (1980–2005) recorded flow rate data of its reference gauge station at Muga stream is taken from Ministry of Water Resource of Ethiopia. After filling gaps and improving the data quality, monthly average flow rate of a year is summarized in Table 3. Year to year variation is observed to occur mainly from rainfall variation in the rainy seasons. Since the determinant flow rate for runoff hydropower system is one available in the dry seasons, the hydro system is less sensitive to year to year variation. A flow rate record of 2004 is taken for having nearly average of the annual flow rates of the latest 5 years.

3.2. Wind potential assessment

Annual average wind speed at a nearby station (Debre Markos) is calculated as 3.5 m/s based on anemometer data collected at 10 m height [6]. NASA has estimated the annual average wind speed of the location to be 3.1 m/s at the same 10 m height [18]. National Metrological Agency (NMA) has changed its anemometer measurement height from 10 m to a data logger measurement at 2 m height. 10 m extrapolation of the wind speed data obtained from the NMA are observed to be 2.4 m/s which is very far below from its previous 10 m measurement [6]. Generally, 10 m height is the one where most standard measurements are taken and measurements at 2 m height are error prone due to vegetation, shading and obstacles in the vicinity. The geographical layout of the studied area is different from the nearby stations where the measurements are taken. The authors of this paper believe that the unevenness nature of the upper Blue Nile gorge is a good resource of wind, although there was no enough time to collect data and prove this. Hence, the minimum of the 3.5 m/s and the 3.1 m/s mentioned above is considered for this study and that is data obtained from NASA. This data can be extrapolated to the selected wind turbine height using Eq. (2) [6,8].

$$v(z)\cdot\ln\left(\frac{z_r}{z_0}\right) = v(z_r)\cdot\ln\left(\frac{z}{z_0}\right) \tag{2}$$

where Z_r is the reference height (m); Z the height where wind speed is to be determined (m); Z_0 the measure of surface roughness (0.1–0.25 for crop land); $v(z)$ the wind speed at height of Z m (m/s) and $v(z_r)$ is the Wind speed at the reference height (m/s).

The value of Z_0 is taken as 0.1; Weibull parameters are estimated to be $K = 2$ and $c = 3.9$ m/s [6]. Table 4 summarizes the wind speed at 10 m and 25 m heights.

3.3. Solar potential assessment

HOMER software requires either monthly average or hourly solar radiation data series. Unfortunately, there is only sunshine duration data recorded at the two nearby stations (Debre Markoss and Motta). However, it is possible to estimate the monthly average solar radiation data using a series of equations [7,8,10,24] which are given by Eqs. (3)–(7).

$$H = H_0\left(a + b\frac{n}{N}\right) \tag{3}$$

where H is the monthly average daily radiation on horizontal surface (MJ/m^2); H_0 the monthly average daily extraterrestrial radiation on a horizontal surface (MJ/m^2); N the maximum possible daily hours of bright sunshine and n is the monthly average daily number of hours of bright sunshine. a and b are regression coefficients having average value of $a = 0.33$ and $b = 0.43$ [7].

Monthly average extraterrestrial radiation is computed using Eq. (4).

$$H_0 = \frac{24*3600*G_{sc}}{\pi}\left(1 + 0.033*\cos\left(\frac{360n_d}{365}\right)\right) * \left(\cos\phi\cos\delta\sin\omega_s + \frac{\pi\omega_s}{180}\sin\phi\sin\delta\right) \tag{4}$$

where n_d is the day number starting from January 1st as 1; G_{SC} the 1367 W/m^2, the solar constant; ϕ the latitude of the location (10.25°); δ the declination angle (°) given by Eq. (5) and ω_s is the Sunset hour angle (°) given by Eq. (6).

$$\delta = 23.45\sin\left(360\frac{248 + n_d}{365}\right) \tag{5}$$

$$\omega_s = \cos^{-1}(-\tan\phi\tan\delta) \tag{6}$$

Table 4
Monthly average wind speed (m/s) at Debre Markoss NASA [18].

	January	February	March	April	May	June	July	August	September	October	November	December	Av.
NASA (10 m)	3.4	3.4	3.2	3.0	3	3.4	3.5	2.9	2.5	2.5	2.9	3.2	3.1
At 25 m	4.1	4.1	3.8	3.6	3.6	4.2	4.1	3.5	3.0	3.0	3.5	3.8	3.7

Table 5
Monthly solar radiation at the project area.

Mid of month	n_d	δ (°)	ω_s (°)	N (hours)	n	H_0 (kWh/m^2/day)	n/N	H (kW/m^2/day)	NASA
January 15	15.00	−21.3	86.01	11.47	9.90	8.82	0.86	5.87	6.15
February 14	45.00	−13.7	87.53	11.67	10.10	9.53	0.87	6.35	6.49
March 15	74.00	−2.89	89.52	11.94	8.40	10.22	0.70	6.57	6.57
April 15	105.00	9.34	91.75	12.23	8.00	10.54	0.65	6.63	6.48
May 15	135.00	18.74	93.56	12.47	8.10	10.45	0.65	6.56	6.35
June 15	166.00	23.30	94.51	12.60	7.30	10.29	0.58	6.18	5.80
July 15	196.00	21.56	94.14	12.55	5.80	10.32	0.46	5.59	5.24
August 15	227.00	13.87	92.60	12.35	4.90	10.44	0.40	5.22	5.26
September 15	258.00	2.33	90.47	12.06	7.90	10.30	0.65	6.48	5.87
October 15	288.00	−9.49	88.31	11.78	9.10	9.74	0.77	6.40	6.28
November 15	319.00	−19.1	86.46	11.53	9.50	8.99	0.82	5.96	6.11
December 15	349.00	−23.3	85.58	11.41	9.70	8.58	0.85	5.71	6.00
Average								6.13	6.05

Table 6
Monthly average of daily primary and deferrable loads for the week days.

Load type category	Demand (kWh/day)					
	January	February–May	June	July and August	September	October–December
Deferrable	312	312	265.2	218.4	265.2	312
Primary						
Household	13093.5	13093.5	13093.5	13093.5	13093.5	13093.5
Health Posts	119.9	119.9	119.9	119.9	119.9	119.9
Primary Schools	13.2	135.6	135.6	13.2	135.6	135.6
Mills	2187.5	2187.5	2187.5	2187.5	2187.5	2187.5
Total	15726.1	15848.5	15801.7	15632.5	15801.7	15848.5

The day length, N, is the maximum possible daily sunshine hour is given by Eq. (7).

$$N = \frac{2}{15}\omega_s \tag{7}$$

Based on the above Eqs. (3)–(7), the solar radiation of the district is estimated as shown in Table 5. The last column of Table 5 shows solar radiation data obtained from NASA in (kWh/m^2/day).

Data from SWERA (Solar and Wind Energy Resource Assessment) shows that the area has a solar radiation potential of 6.4 kWh/m^2/day. For this study the calculated annual average data 6.13 kWh/m^2/day is considered.

4. Load estimation

According to Bekele et al. [25], electric load in the rural villages of Ethiopia can be assumed to be composed of lighting, radio and television, water pumps, health post and primary schools load. Tamrat [10] considered only lighting, radio and television as a community load. In this study, electricity for cooking and for flour mills is added to the load together with home radio and a TV set. Water pumps are considered as deferrable loads while the others as primary loads. As indicated previously, there are about 63,000 people without electricity now or even in the near future. Assuming an average of six members in a family, there would be a total of 10,500 families. Assuming one elementary school and one health center per 420 families, a total of 25 primary schools and 25 health centers are required for the community.

Each primary school consists of eight classrooms and the classroom will be installed with four 11 W CFLs (compact fluorescent lamps) and a radio receiver. An 11 W CFL has an equivalent lumens effect of a 60 W incandescent lamp. Additional four CFLs (of 11 W) for external lighting are also considered. Evening classes are conducted from 18:00 to 21:00 and the receivers are used for radio lessons. Similarly for the 25 health centers, having three rooms each, one 11 W CFL per room and one 20 W CFL for external lighting are considered. A vaccine refrigerator of 80 W working for 24 h; a 20 W capacity microscope and a 5 W radio receiver for the office hours; and a 1 kW water heater operational for 3 h per day are also suggested for each health centers. One ceiling fan (75 W) per room is also to be installed for air conditioning purpose between 10:00 and 15:00.

For the community a total of 25 flour mills of 12.5 kW working from 9:00 to 12:00 and 14:00 to 18:00 are assumed. Each family is assumed to use electric stove of 3 kW rating for baking a local bread, “Injera”, for 40 min every fourth day. The timing for baking is assumed to be in either of 6:00–9:00, 11:00–14:00 or 15:00–18:00. Each household is to be installed with three CFLs of 11 W rating to be lit from 18:00 to 23:00. Additionally a CFL of 11 W is also considered for external lighting. A radio receiver (5 W) and a TV set (70 W) are to be used in the time 18:00–23:00.

Water pumping system is required for the households, the schools and health care centers. A minimum of 100 l of water per day per family and 2400 l/day for each pair of one health center and one primary school is suggested [8,25]. To accomplish this, 110 pumps of 450 W (with a capacity of 30 l/m) operating for

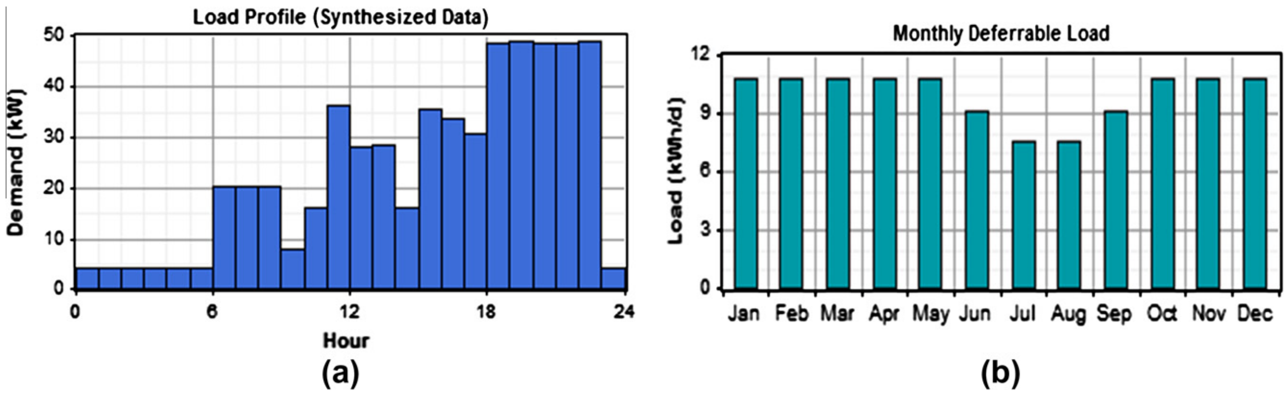

Fig. 4. (a) Daily primary load profile and (b) monthly variation of deferrable load.

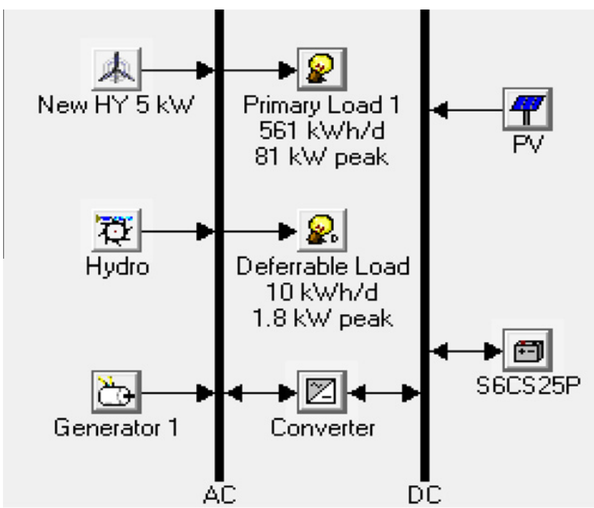

Fig. 5. Hydro/PV/Wind hybrid setup.

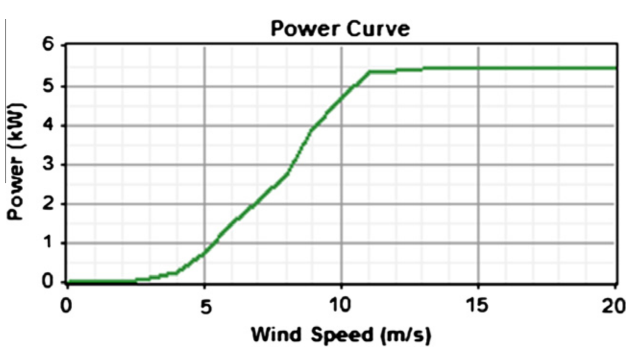

Fig. 6. Power curve of HY-5 kW wind turbine.

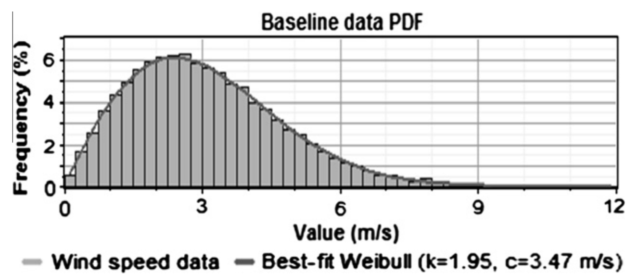

Fig. 7. Probability density function of wind speed (at 10 m height).

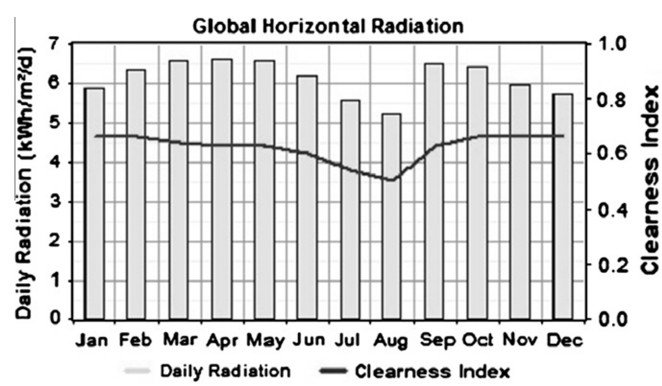

Fig. 8. Monthly solar radiation (kW/m^2).

6 h/day are to be installed to supply water for the community. Another 25 pumps of 150 W (with a capacity of 10 l/m) for the 25 schools and health centers operating for 4 h/day are assumed. A water storage capacity of 4 days is suggested requiring a storage capacity of 1188 kWh for the community and 60 kWh for primary school and health centers. The peak deferrable load is 3.75 kW for primary school and health center and 49.5 kW for the community.

In the weekend, mills are not working and evening classes are conducted at day time. TV and radio are used from 10:00 to 17:00. Up to 30% deferrable load decreases can be expected in the rainy season [25]. 15% load decrease for June and September while 30% for July and August are assumed. There are no classes in July and August (annual break) and in January (semester break). During these times, only external lightings would be operational.

Based on the above analysis, the community will have a primary peak demand of 1424 kW, a deferrable peak demand of 52.25 kW and a storage capacity of 1248 kWh. The variation of the deferrable load and the daily primary load (for week days) throughout the year is shown in Table 6.

The main objective of the overall project is to propose the optimal interconnection of the six small hydropower potentials together with wind and PV systems in a form of a hybrid system, which will improve the system reliability and investment costs. But, there is a limitation of the HOMER software to handle more than one hydro resource at the same time. Hence, modular analysis is opted for and the nominal hydropower potential of each

Table 7
Inputs to HOMER software.

	PV module	Wind turbine	Hydro system	Diesel generator	Battery surrette6cs25p	Converter
Size (kW)	1	5	34.2	0–77	1156 Ah	1
Capital ($)	1800–3000	4990	70,000	0–9000	833	700
Replacement cost ($)	1800–3000	3327	10,000	0–6000	555	700
O & M cost ($/yr)	25	100	100	0.4/hr	15	10
Sizes (kW) considered	0, 5, 10, 15, 20, 25, 30, 35, 40	–	34.2	0, 22, 26, 33, 35, 44, 53, 77	–	0, 10, 20, 30, 40, 50
Quantities	–	0, 1, 2, 3, 4, 5	–	–	0, 20, 40, 60, 80, 100, 120, 140	–
Life time	20 yrs	20 yrs	>25 yrs	40,000 h	9645 kWh	15 yr

Table 8
Over all optimization results corresponding to the current sensitivity values.

Rank	PV (kW)	HY-5 kW	Hydro (kW)	Gen (kW)	S6CS25P	Converter (kW)	Dispatch strategy	Total NPC ($)	COE ($/kWh)	Renewable fraction	Capacity shortage	Diesel (L)	Gen (hrs)
1	0	0	34.2	22	40	30	LF	246,787	0.101	0.9	0.04	6783	1491
2	0	1	34.2	22	40	30	LF	249,586	0.102	0.9	0.04	6479	1440
3	0	0	34.2	22	40	20	LF	250,773	0.103	0.88	0.03	8049	1853
4	0	0	34.2	22	60	30	LF	251,167	0.103	0.92	0.03	5257	1175
7	5	0	34.2	22	40	30	LF	253,051	0.103	0.91	0.03	6058	1355
–	–	–	–	–	–	–	–	–	–	–	–	–	–
19	0	2	34.2	22	60	30	LF	258,229	0.105	0.93	0.02	4763	1080
20	5	0	34.2	22	60	40	LF	258,757	0.106	0.94	0.03	3857	835
–	–	–	–	–	–	–	–	–	–	–	–	–	–
48	5	2	34.2	22	60	40	LF	264,786	0.108	0.95	0.02	3283	715
49	0	1	34.2	33	40	30	LF	265,372	0.107	0.88	0.01	7986	1258
–	–	–	–	–	–	–	–	–	–	–	–	–	–
116	15	2	34.2	0	80	40	CC	273,650	0.114	1	0.05	0	0
117	15	2	34.2	0	80	40	LF	273,650	0.114	1	0.05	0	0
–	––	–	–	–	–	–	–	–	–	–	–	–	–
140	10	0	34.2	22	60	50	LF	275,407	0.112	0.95	0.02	3163	682
141	20	0	34.2	0	80	40	CC	275,440	0.115	1	0.05	0	0
142	20	0	34.2	0	80	40	LF	275,440	0.115	1	0.05	0	0
–	–	–	–	–	–	–	–	–	–	–	–	–	–
199	0	2	34.2	44	40	30	LF	279,416	0.112	0.88	0	8775	1083
200	5	0	34.2	33	60	50	LF	279,446	0.112	0.93	0.01	4771	732

Table 9
optimization results in a categorized form.

Rank	PV (kW)	HY-5	Hydro (kW)	Gen (kW)	S6CS25P	Converter (kW)	Dispatch strategy	Total NPC	COE ($/kWh)	Renewable fraction	Capacity shortage	Diesel (L)	Gen (hrs)
1	0	0	34.2	22	40	30	LF	$246,787	0.101	0.9	0.04	6783	1491
2	0	1	34.2	22	40	30	LF	$249,586	0.102	0.9	0.04	6479	1440
3	5	0	34.2	22	40	30	LF	$253,051	0.103	0.91	0.03	6058	1355
4	5	1	34.2	22	40	30	LF	$255,568	0.104	0.92	0.03	5743	1291
5	15	2	34.2	0	80	40	CC	$273,650	0.114	1	0.05	0	0
6	20	0	34.2	0	80	40	CC	$275,440	0.115	1	0.05	0	0
7	0	0	34.2	33	0	0	LF	$368,348	0.149	0.72	0.03	23,646	3703
8	0	1	34.2	33	0	0	LF	$371,807	0.151	0.72	0.03	23,387	3672
9	5	0	34.2	33	0	10	LF	$380,528	0.154	0.73	0.02	22,579	3539
10	5	1	34.2	33	0	10	LF	$383,954	0.155	0.74	0.02	22,318	3507

site is determined as a percentage of the sum total. Based on the percentage contribution, the primary and the deferrable loads are allocated to each site. A hybrid system at each site can be then analyzed and finally the results will be combined into single interconnected system. Muga1, Muga2, Suha, Bechet, Taba (B) and Taba (A) share the community load in the 21.7%, 56.3%, 7.5%, 8.3%, 3.45% and 2.35%, respectively. Taba (B) with a 3.45% load sharing is selected for further analysis in this paper. The primary and the deferrable load profiles at Taba (B) are shown in Fig. 4a and b.

5. System setup and inputs to HOMER for Taba (B)

Components of the hybrid system at Taba (B) consists of HY-5 kW wind turbine, PV module, hydro generator, converter, battery, diesel generator, primary and deferrable loads. For random variation of primary load 15% daily and hourly noise level is added. Fig. 5 shows HOMER generated hybrid setup. Fig. 6 is the power curve for the selected HY-5 kW wind turbine [26]. The synthesized wind speed PDF and solar radiation data is shown in Figs. 7 and 8 respectively.

Table 10
System report for 95% renewable fraction.

System architecture		Sensitivity case		Annual electric production (kWh/yr)			Annual electric consumption (kWh/yr)			Emission (kg/yr)	
PV	5 kW	Design flow rate	75 l/s	PV array	10,226	4%	AC primary load	201,689	98%	CO_2	8646
Wind turbine	2 HY5 kW	Stream flow rate	1540 l/s	Wind turbine	7956	3%	Deferrable load	3632	2%	CO	21.3
Hydro	34.2 kW	PV capital cost multiplier	0.8	Hydro turbine	232,077	88%	Total	205,321	100%	Unburned CH_4	2.36
Gen	22 kW	wind data	3.1 m/s	Gen	12,907	5%	Cost summary			Particulate matter	1.61
Battery	60 Surr6CS25P	PV capital cost multiplier	0.8	Excess electricity	38,291	14.6%	Total NPC	$264,786		SO_2	17.4
Inverter	40 kW	Solar data	6.12 kWh/m^2/d	Unmet load	3077	1.5%	Levelized COE	$0.108/kWh		NO_x	190
Rectifier	40 kW			Capacity shortage	4725	2.30%	Operating cost	$7384/yr			
Disp. strategy	LF			Renewable ratio	0.95	95%					

Table 11
Monthly averaged electric production (kW).

Month	January	February	March	April	May	June	July	August	September	October	November	December
PV	1.207	1.268	1.259	1.215	1.163	1.081	0.99	0.945	1.217	1.259	1.225	1.186
Wind	1.183	1.186	0.997	0.828	0.828	1.281	1.186	0.747	0.462	0.464	0.747	1
Diesel generator	2.5965	7.6047	2.1067	0.8044	0.6358	0.5762	0.4976	0.9929	0.7459	0.5435	0.4064	0.6559
Hydro	20.164	11.246	23.654	29.08	29.08	29.08	29.08	29.08	29.08	29.08	29.08	29.08

Table 7 shows the summary of the data input to HOMER. Some of the components are expressed in kW while others in quantities. Cost information of small hydropower, solar PV module, battery, converter, diesel generator and wind turbine is collected from local suppliers, manufacturers and from different websites [26–30]. The life time of battery is indicated by its maximum throughput.

5.1. Additional inputs and sensitivity variables

In addition to the inputs summarized in Table 7, sensitivity variables are defined for diesel cost range of $0.6–$1.2/l and PV capital and replacement cost multipliers of 0.6–1 are used for sensitivity analysis. Inverter and converter efficiencies are assumed to be 90%. Interest rate of 6.7% is used for present cost analysis. To account greenhouse effect, a 20/t of penalty for CO_2 emission is considered. Both cyclic charge (CC) and load follow (LF) dispatch strategies are analyzed. The design flow rate is repeatedly simulated for a range of 30 l/s to 100 l/s and optimal design flow rate of 75 l/s is obtained. A 10 l/s reserve flow rate is considered. The hydropower system has a gross head of 62 m and a pipe head loss of 15% is assumed. Hydropower efficiency of 75% with 35% minimum flow ratio and 100% maximum flow ratio is taken. Derating factor of 90% and 20% ground reflectance PV system without tracking system is considered. PV panels are to be mounted at slope of 10° (latitude of the site). An operating reserve of 10%, 25% and 50% of the hourly load, solar power output and wind power output respectively is suggested. The fuel curve characteristics are calculated using HOMER and an intercept coefficient of 0.02 l/hr/kW and 0.23 l/hr/kW are found using data from manufacturer's website [30]. Diesel generator is allowed to operate under a minimum load ratio of 70%. Maximum annual energy shortage and minimum renewable fraction are set to 5% and 0% respectively.

6. Results and discussion

HOMER simulates all the possible system configurations based on the combinations of the components specified to it as input data and discards the infeasible system configurations that do not adequately meet the suggested load, the available resource and/or specified constraints [31]. Hence, only feasible combinations are displayed according to the total net present cost (NPC) in an increasing order. The optimization results are given out in an overall form (Table 8) and in a categorized form (Table 9). For a particular set of sensitivity variables (solar radiation, average wind speed, diesel price, etc.), the overall table displays all feasible system configurations according to cost effectiveness. Table 8 shows selected few system configurations from the top 200 listed in the truncated long overall table. The categorized table displays only the most cost effective configuration from each possible combination of components.

It is to be noted that the results can further be refined with the refinement of the component sizes, but at a cost of much longer running time of the software. In this work a step by step repeated simulation is carried out by varying the input variables from coarse to fine and a practically applicable results have been achieved (Tables 8 and 9).

Since the price for diesel and for PV panels are more dynamic than other types of components, a range of diesel price and a PV

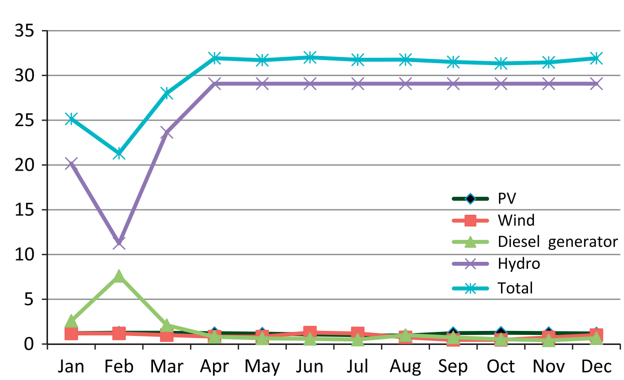

Fig. 9. Monthly averaged electric production (kW).

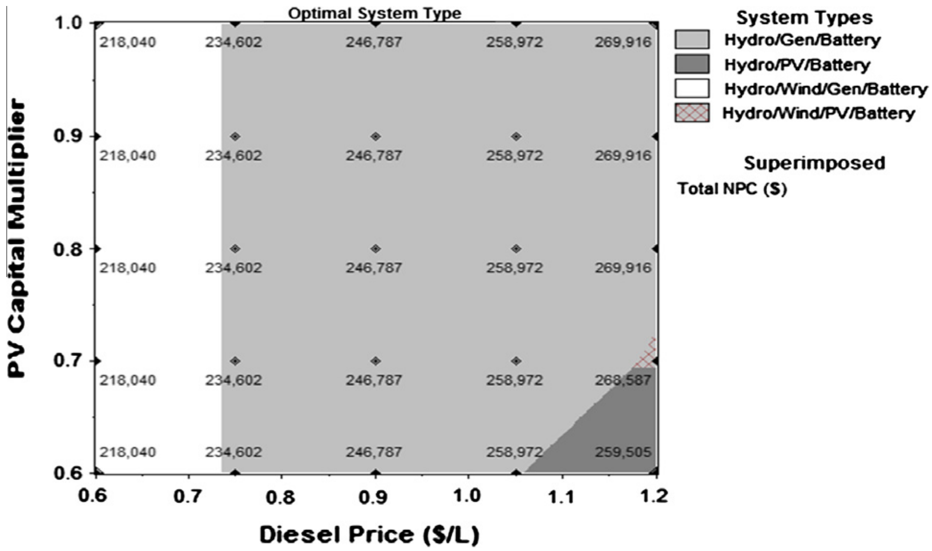

Fig. 10. Sensitivity analysis for diesel and PV module price variation.

capital cost multipliers are used as sensitivity parameters. HOMER displays the sensitivity analysis both in tabular and graphical form.

The results obtained are based on the current price of diesel ($0.9/l) and PV capital of $2400/kW. From Tables 8 and 9, the levelized COE is observed to be in the range of $0.10–$0.16/kWh. Load follow (LF) dispatch strategy dominates more than cyclic charge (CC). Presence of plenty of renewable resource is indicated by the level of the renewable fraction ranging from 72% to 100% in the renewable fractions column of the Tables 8 and 9. Capacity shortage ranges from 0% to 5%. There are feasible combinations without any diesel generator (a renewable fraction of 100%). There are also system types which do not require battery and converter. It can be seen from the two tables, that the Hydro/Diesel/Battery system is the most cost effective with a 90% of renewable fraction and 4% capacity shortage. This system type is followed by Wind/Hydro/Diesel/Battery system with a slightly higher total NPC.

The current energy tariff of Ethiopia is below $0.04/kWh. A previous study of Wind/PV hybrid system in Ethiopia (without hydropower resource) [25] has suggested an optimal system with a levelized COE of $0.3–$0.4/kWh. Global electricity tariff varies in a wide range. For instance Canada has an energy tariff of $0.0618/kWh while Denmark has $0.43/kWh [32]. All feasible system types shown in both tables here have a levelized COE slightly higher than the national tariff ($0.101–$0.115/kWh). However, they are within the range of global tariff and much lower than what a previously studied result showed.

The listed outputs of the overall optimization table (Table 8) have little variation in the COE ($0.101–$0.16/kWh). A system setup ranked as 48th has renewable fraction of 95% and a capacity shortage of 2% for an additional COE of $0.007/kWh over the top ranked system. This can be a good choice for implementation. The system report for this particular setup is summarized in Table 10. Also, the monthly electricity production of each component is given in Table 11 with its plot shown in Fig. 9.

Sensitivity analysis is also carried out and Fig. 10 shows the variation of PV capital cost multiplier against diesel price.

As it can be seen from Fig. 10, Wind/Hydro/Generator/Battery system is favored as diesel price decreases. The most economical (least NPC) system is insensitive to PV capital multiplier at low diesel prices. As diesel price increases beyond some $0.74/l, the Hydro/Generator/Battery system dominates overtakes. When diesel price still continues to increase and surpasses the $1.05/l limit, solar resources will be favored and the system starts to be sensitive to PV capital multiplier. For further increase of the diesel price, the Hydro/PV/Wind/Battery system becomes cost effective.

7. Conclusion

In this paper, detailed study of Taba (B) site is carried out. GIS based hydrological and empirical estimations are used to determine the flow rate at the site and it is found to be in the range of 39–7666 l/s with a mean annual flow rate of 1540 l/s. A 62 m gross head and a 75 l/s design flow rate giving a nominal hydropower potential of 34.2 kW are considered. Monthly average wind speed data from NASA is used to synthesize hourly wind speed data using HOMER. The Weibull parameters k and c are determined ($k = 1.95$ and $c = 3.47$ m/s). Solar radiation is calculated from daily sunshine hour data using empirical formulas and is 6.13 kWh/m^2/day. This result is very close to what is obtained in the previous study, and also to NASA and SWERA predictions.

Hourly electric load of the community consisting of lighting, TV set, radio receiver, stove, health post, clinic, water pumps and flour mills is determined. The total community (about 10,500 families) is estimated to have a primary peak demand of 1424 kW, a deferrable peak demand of 52.25 kW and a storage capacity of 1248 kWh. A 3.45% share of the total load for Taba (B) as per its resource is considered for analysis using HOMER. Diesel price and PV capital and replacement cost are used for sensitivity analysis.

In the overall results, different optimum and feasible system configurations with different level of renewable fraction and total NPC are obtained. The levelized COE ranges from 0.1/kWh to 0.121/kWh. This cost is slightly higher than the current energy tariff within the country (<$0.04/kWh), but, is much less than previously studied PV/Wind based hybrid system (does not include hydro). It can be said the COE determined here is at the lower range of global electricity tariff.

According to the total NPC for the current diesel price of $0.9/l and a PV capital of $2400/kW, Hydro/Generator/Battery, Hydro/Wind/Generator/Battery, Hydro/PV/Generator/Battery, Hydro/Wind/PV/Generator/Battery, Hydro/Wind/PV/Battery and Hydro/PV/Battery systems are ranked in descending order as feasible options. From the sensitivity analysis, a diesel price of less than $0.74/l is observed to favor Hydro/Wind/Generator/Battery systems while diesel price of $0.74–$1.05/l favors Hydro/Generator/Battery system. For diesel prices higher than $1.05/l, Hydro/PV/Battery and

Hydro/Wind/PV/Battery are better options at a lower and higher PV capital cost multipliers respectively.

Although the proposed system has a relatively higher COE than the national tariff, in view of the energy shortage at the national level, resistance to deforestation, clean energy development, changing the life of the poor in remote regions and expansion of power generation it is a highly commendable solution.

The hydropower system suggested is a run-off type and does not require extensive civil work. The head is naturally endowed. Consequently, all the civil structures and installation of electromechanical components are to be realized within a small area, which as a result would not cause any significant social and environmental impacts.

This study, even though is specific for Taba (B) site, it proves to be of high importance in its demonstrative character to the utilization of the resource potential of the region. As this is just a part of an ongoing project, the research will proceed for the remaining sites so that the 10,500 families in the area will be provided with modern electricity. Ethiopia has a lot of small hydropower potential. Due to the seasonal nature of these resources, wind and solar resources together with diesel generator and battery system as a backup can be used for better reliability of rural electrification.

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